EDA & UNSUPERVISED LEARNING · CHAPTER II

Bivariate Analysis

How to measure — and see — the relationship between two columns

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THE WHOLE CHAPTER ON ONE SLIDE

Pick the tool by the type of the two columns

Pair of columns	Indicator (a number)	Graph (a picture)
Quantitative vs Quantitative	Covariance, Pearson r, Spearman ρ	Scatterplot
Quantitative vs Categorical	η^2 (eta-squared)	Box / violin plot per group
Categorical vs Categorical	χ^2 test, Cramér's V	Stacked / grouped bars, treemap

Ordinal columns (e.g. Likert scales) can use Spearman ρ, because it works on ranks.

Running example: the Gapminder dataset

1,704 rows × 5 columns: country, continent, lifeExp, pop, gdpPercap.

Every 5 years from **1952 to 2007** — most examples zoom in on the year 2007.

Pop and GDP have means far above their medians: a few huge values dominate, so log scales help later.

2007	Mean	Median	Min	Max
pop	44.0 M	10.5 M	0.2 M	1,318.7 M
lifeExp	67.0	71.9	39.6	82.6
gdpPercap	11,680	6,124	278	49,357

2.1 · QUANTITATIVE VS QUANTITATIVE

Covariance: the direction of a linear trend

$$\mathrm{Cov}(X, Y) = \frac{1}{n-1} \cdot \sum (x_i - \bar{x})(y_i - \bar{y})$$

Positive

X and Y tend to move in the **same** direction.

Negative

X and Y tend to move in **opposite** directions.

The catch

Its size depends on the **units/scale** of X and Y, so the value itself is hard to interpret.

Pearson correlation r: covariance without units

$$r = \mathrm{Cov}(X, Y) / (s_X \cdot s_Y),\ -1 \le r \le 1$$

- **r ≈ 1**: positive linear correlation (upward trend)
- **r ≈ −1**: negative linear correlation (downward trend)
- **r ≈ 0**: no linear trend — or a trend hidden by noise
- Use it to find **useful inputs** and **redundant inputs** for linear models

⚠ **Correlation shows a tendency, not cause and effect.**

TIPS EXAMPLE

0.676

Corr(tip, total bill): bigger bills come with bigger tips.

2.1 · CORRELATION MATRIX

Many columns at once: the correlation matrix

Pearson (Gapminder 2007)

	gdp	lifeExp	pop
gdp	1	0.68	−0.06
lifeExp	0.68	1	0.05
pop	−0.06	0.05	1

Spearman (Gapminder 2007)

	gdp	lifeExp	pop
gdp	1	0.86	−0.06
lifeExp	0.86	1	0.00
pop	−0.06	0.00	1

Read it: GDP and life expectancy rise together — and the link is even stronger as a monotonic (curved) trend. Population looks unrelated to both.

2.2 · QUANTITATIVE / ORDINAL

Spearman ρ: correlation on ranks

Pearson is sensitive to outliers, sees only **linear** links, and doesn't suit ordinal data.

Spearman replaces each value by its **rank**, then measures the **monotonic** relationship.

$$\rho = 1 - 6 \Sigma d_i^2 / (n(n^2 - 1))$$

d_i = rank of x_i – rank of y_i. ρ is large when the ranks mostly match.

Worked example

```
X = [1, 3, 2, 5, 4]
Y = [0.1, 1, 0.2, 1000, 1.5]
```

Ranks are identical → every $d_i = 0$ → **ρ = 1**.

Pearson gives only **r = 0.71**, dragged down by the outlier 1000.

2.2 · SUMMARY

Pearson vs Spearman

Aspect	Pearson r	Spearman ρ
Type	Parametric	Non-parametric
Measures	Linear relationship	Monotonic relationship
Data	Numerical	Numerical or ordinal
Outliers	Sensitive	Less sensitive
Range	[−1, 1]	[−1, 1]
Value ≈ 0	No linear trend (or noise)	No monotonic trend

2.1 · VISUALIZATION

Scatterplot: see the trend

GDP vs Life expectancy

Clear upward trend: richer countries tend to be healthier.

But a few countries with above-average economies still have poor health.

GDP vs Population

No clear trend.

GDP per capita does not seem to depend on how many people a country has.

Tip: use a **log scale** when points are crowded in one corner — and say so clearly on the axis, so nobody misreads it.

η^2: how much the groups explain

$$BSS = \sum_g n_g (\bar{x}_g - x)^2$$

$$TSS = \sum_i (x_i - x)^2$$

$$\eta^2 = BSS / TSS,\ 0 \le \eta^2 \le 1$$

BSS: how far apart the group means are. TSS: the total spread of X.

X and group G are related when X is **well separated** between groups and **not too spread** inside each group.

$\eta^2 \approx 0$: no relation · **$\eta^2 \approx 1$**: strong relation.

It is the core of ANOVA — and, like Pearson, sensitive to outliers.

2.3 · RESULT AND VISUALIZATION

Continent explains life expectancy more than GDP

H² · LIFEEXP VS CONTINENT

0.635

H² · GDPPERCAP VS CONTINENT

0.424

- **Box / violin plot per group** is the go-to chart: one box per continent.
- Conditional histograms or densities also work, but are used less often.
- 🔑 Clearly separated groups = a connection between the two columns.

2.4 · CATEGORICAL VS CATEGORICAL

Compare category mixes across groups

GDP is cut into three equal-sized classes at the 33% and 67% quantiles: **Developing · Emerging · Developed**.

Treemap / mosaic

Blocks sized by share, nested by continent then GDP class.

Stacked bars

Percent of each GDP class inside each continent.

Grouped bars

Counts side by side for each continent.

Read it: the more the bars differ between groups, the stronger the association — African countries are mostly Developing, European ones mostly Developed.

2.4 · INDICATOR

χ^2 test of independence

Hypotheses

H_0: X is independent of Y

H_1: X is not independent of Y

If H_0 is true, observed ≈ expected frequencies in every cell.

1. Build the contingency table of counts n_{ij} (I rows × J columns).
2. Expected counts under H_0: $e_{ij} = n_{i\cdot} \times n_{\cdot j} / N$.
3. Distance: $\chi^2 = \Sigma\ (n_{ij} - e_{ij})^2 / e_{ij}$.
4. Under H_0, χ^2 follows $\chi^2(df)$ with $df = (I-1)(J-1)$.
5. p-value = $P(\chi^2(df) \geq \text{observed } \chi^2)$.

p-value < 0.05 → reject H_0: the variables are associated.

2.4 · EXAMPLE

France, 2021: job vs support for the Covid vaccine

Favor	F	SE	SEP	MEP	E	MW	NW
3 (strong)	263	1085	7704	4346	4052	1551	2319
2	616	2265	10088	8889	11264	4713	3892
1	222	950	2864	3238	4517	1910	1145
0 (weak)	126	562	1105	1420	2596	1531	622

df = 3 × 6 = 18, $\chi^2 \approx 3298$ — far beyond what chance gives, so job and vaccine attitude are clearly **not independent**.

F farmers · SE self-employed · SEP senior executives · MEP middle executives · E employees · MW manual workers · NW never worked / others. Survey of 85,855 people.

2.4 · STRENGTH OF ASSOCIATION

Cramér's V: χ^2 rescaled to [0, 1]

$$V = \sqrt{\left(\chi^2/n\right) / \min(I-1, J-1)}$$

V ≈ 0: no association · **V ≈ 1**: strong association.

χ^2 tells you *whether* there is a link; V tells you *how strong* it is.

Bias correction

Plain V is a biased estimator and can overestimate the strength. Corrected version:

$$\tilde{\varphi}^2 = \max\left(0, \chi^2/n - df/(n-1)\right)$$
$$\tilde{I} = I - (I-1)^2/(n-1)$$
$$\tilde{J} = J - (J-1)^2/(n-1)$$
$$\tilde{V} = \sqrt{\tilde{\varphi}^2 / \min(\tilde{I}-1, \tilde{J}-1)}$$

2.5 · MULTIPLE INFORMATION

Add more variables to one scatterplot

Color

Categorical → distinct colors (continent).

Quantitative → a gradient (log population).

Shape / symbol

Categorical only (GDP class).

Size

Quantitative (population) → a **bubble chart**. Pick a sensible maximum size.

Careful: many colors plus many symbols quickly make a graph unreadable. Add a channel only when it helps the message.

2.6 – 2.8 · TIME

When one of the variables is time

Line plot

A quantitative column over the years.

Example: lifeExp, pop and GDP of Oceania countries, 1952–2007.

Evolution barplot

A categorical column over the years: one stacked bar per year.

Example: GDP classes of Asian countries.

Animated graph

A scatterplot with one frame per year (Plotly *animation_frame*).

Example: GDP vs lifeExp, size = pop, color = continent.

GOOD PRACTICE

Seven keys to a proper visualization

🎯 **Clear purpose**
Distribution, relationship or comparison?

👥 **Fit the audience**
Experts, public, decision-makers, students

🧠 **Clear and honest**
No distortion; label any log scale

♿ **Accessible**
Color-blind-safe colors, contrast, alt text

📊 **Right chart**
Bar, line, box, heatmap, scatter…

🎨 **Clean and consistent**
Clear labels, titles, legends, font size

⚙️ **Interaction if needed**
Tooltips, zoom, filters, animation

TAKEAWAY

Begin with the indicators.
Then visualize the pairs that matter.

Quan–Quan: r, ρ + scatter · Quan–Qual: η^2 + boxplots · Qual–Qual: χ^2, Cramér's V + bars